Model Comparison Report – Subtask 1 (Polarization Detection)

Experimental Setup

Two models were evaluated on the development set:

1. **Baseline Model**: TF-IDF vectorization + Logistic Regression
2. **Transformer Model**: XLM-RoBERTa fine-tuned for binary classification

Evaluation metric: **Macro F1 score**

Results

Model	Macro F1
TF-IDF + Logistic Regression	0.7020
XLM-RoBERTa (fine-tuned)	**0.7606**

The transformer model achieved a **8.35 percentage point improvement** over the baseline.

Analysis

The transformer significantly outperformed the baseline model. This improvement can be attributed to:

- Pretrained multilingual contextual embeddings
- Ability to model long-range dependencies
- Better semantic representation of polarized language

The baseline model relies on surface-level lexical features (n-grams), which limits its ability to detect subtle forms of polarization.

Transformer Model

Overall Performance Interpretation

Confusion matrix:

- TP = 52
- FN = 7
- TN = 95
- FP = 6

Total = 160
Errors = 13
Accuracy ≈ 91.9%
Macro F1 ≈ 0.776

What this tells us

- Errors are balanced (7 FN vs 6 FP)
- No extreme bias toward one class
- The model separates classes fairly well
- Mistakes are not random - they're systematic

This is a healthy confusion matrix.

What Kind of Errors Is the Transformer Making?

Pattern A: Sarcasm & Irony

Examples:

- "should run for gop press. lol"
- "we build a wall around the red states and cele..."
- "i guess steve didnt get the memo…"

These are sarcastic or mocking.

Transformer failure reason:
XLM-RoBERTa struggles with pragmatic tone detection when context is missing.

Sarcasm is hard without conversational history.

Pattern B: Implicit Polarization (No Explicit Attack)

Examples:

- “who gave the order to stop counting votes…”
- “mayorkas secretly met with sorosfunded groups…”

These imply conspiracy or political framing but are phrased as questions.

The model likely learned:

- Strong attack words => polarization
- Question format => neutral

So it misses *implicit framing*.

Pattern C: Entity Bias

Look at high-confidence wrong examples:

- MSNBC
- Soros
- Mike Pence
- Hamas

The model reacts strongly to named entities.

Example:

> “mike pence is such a nerd”
> True label = 0
> Predicted = 1 (0.93 confidence)

This suggests:
The model associates political names with polarization automatically.

That’s lexical shortcut learning.

Pattern D: Border / Immigration Topics

Several errors involve:

- Open borders
- Asylum seekers
- Immigration

Likely training data has strong polarization patterns tied to these keywords.

Model learned topic => label shortcut.

High-Confidence Errors = Most Important Insight

I have many >0.9 confidence wrong predictions.

That means:

- The model is not confused
- It is confidently biased

It suggests:
The model relies heavily on trigger tokens.

Low-Confidence Correct Cases

Examples:

- “this is a massive scandal”
- “ballot stuffing by dems…”

These are clearly polarized but model confidence is ~0.55.

This means:

The model recognizes signal but is uncertain.

Likely because:

- Training examples were inconsistent
- Label noise exists
- Borderline examples exist in dataset

What the Transformer Is Good At

Based on confusion matrix:

- Clearly explicit polarized statements
- Strong emotional language
- Direct accusations

Transformers excel at semantic richness.

What It Is Bad At

From the data:

1. Sarcasm
2. Implicit framing
3. Short mocking phrases
4. Topic-trigger bias
5. Entity-driven assumptions
